Modular flow generates generalized entropy in type-II_∞ crossed-product gravity: an Araki-cocycle derivation, with a no-go for HPS–DEHK Stinespring intertwining

Kévin Remondière

Independent researcher, Tarbes, France

Submitted to Letters in Mathematical Physics

Preprint: May 5, 2026

Abstract

Working in the type-II_∞ crossed-product algebra $\mathcal{M}_R = \mathcal{A}_{\mathrm{III}_1} \rtimes_{\sigma^\phi} \mathbb{R}$ attached to a static-patch observer (Chandrasekaran–Longo–Penington–Witten, de Vuyst–Eccles–Höhn–Kirklin), we use the Araki relative modular operator $\Delta_{\rho|\sigma}$ and its cocycle identity to derive the algebraic identity

$$\frac{d}{d\tau_R} S_{\mathrm{gen}}(\rho)\Big|_{\rho \text{ fixed},\, \sigma_{\tau_R}} = \langle K_R \rangle_\rho,$$

where K_R is the dressed modular Hamiltonian and $\sigma_{\tau_R} = \sigma \circ \sigma^\sigma_{-\tau_R}$ is the modular-time-evolved KMS reference state. The derivation is rigorous in the type-II_∞ setting via Pusz–Woronowicz applied to normal semifinite faithful (nsf) weights and is independent of the Petz recovery construction (which has no known type-II_∞ realization for $\mathcal{M}_R$).

We then examine, and refute at the operator level, the natural attempt to intertwine K_R with the chord (transfer-matrix) Hamiltonian H_{grav} of the double-scaled SYK / sine-dilaton gravity model (Heller–Papalini–Schuhmann; Heller–Ori–Papalini–Schuhmann–Wang) through a modular-rigidified Stinespring isometry $V : \mathcal{H}_{\mathrm{HPS}} \to \mathcal{H}_{\mathrm{DEHK}}$. The HPS chord spectrum $E(\theta) = -\cos\theta/\sqrt{2|\log q|(1-q^2)}$ is bounded continuous on $[-E_{\max}, +E_{\max}]$, while $K_R = \beta_{\mathrm{dS}} H_\xi + p$ is unbounded continuous on $\mathbb{R}$. No isometry can intertwine a bounded self-adjoint operator with an unbounded one; hence the proposed bridge fails as an operator-level identification.

What does survive is a thermodynamic / Krylov-asymptotic match: the semiclassical large-N partition functions agree, $\log Z_{\mathrm{chord}}(\beta_{\mathrm{dS}}) = \log Z_{\mathrm{DEHK}}(\beta_{\mathrm{dS}}) + O(1/N)$, reproducing the standard Bekenstein–Hawking de Sitter entropy $A_{\mathrm{horizon}}/(4G_N)$. We frame this paper as a *consistency note* (MVT2-class): the Araki theorem stands as a new result of independent interest, while the Stinespring bridge between the DSSYK chord algebra and the de Sitter crossed product is recorded as a no-go obstruction for future work.

1 Introduction

The modular flow / generalized entropy programme. Algebraic methods have transformed the discussion of generalized entropy in gravitational subregions. After Witten's observation [1] that gravitational backreaction promotes the type-III_1 algebra of bulk quantum field theory observables to a type-II crossed product, two parallel programmes have emerged: (i) the de Sitter / static-patch programme of Chandrasekaran–Longo–Penington–Witten [2] (CLPW), refined by de Vuyst–Eccles–Höhn–Kirklin [3] (DEHK), in which the crossed-product trace yields finite observer-dependent entropies, and (ii) the generalized second law (GSL) programme of Faulkner–Speranza [4] (FS24), in which monotonicity of relative entropy under modular inclusions translates to GSL-type statements for arbitrary cuts of Killing horizons.

Both programmes share a central structural object: the *modular Hamiltonian* K_R of the dressed (crossed-product) algebra. In the KMS-thermal de Sitter setting at $\beta_{\mathrm{dS}} = 2\pi r_{\mathrm{dS}}$ (Bisognano–Wichmann [14]; Connes–Rovelli thermal-time hypothesis [15]), K_R generates a genuine one-parameter automorphism group, and its expectation value $\langle K_R \rangle_\rho$ in a perturbed state ρ controls the first-law variation of the dressed entropy.

Two questions this note answers.

1. Does the Araki relative modular operator $\Delta_{\rho|\sigma}$ [7] provide a clean type-II_∞ derivation of $dS_{\mathrm{gen}}/d\tau_R = \langle K_R \rangle_\rho$, without invoking Petz recovery? *Yes* (Theorem 1).

2. Can the chord Hamiltonian H_{grav} of the DSSYK / sine-dilaton holographic dictionary of Heller–Papalini–Schuhmann [5] (HPS), refined by Heller–Ori–Papalini–Schuhmann–Wang [6] (HOPSW), be intertwined with K_R via a CPTP isometry between the chord Hilbert space $\mathcal{H}_{\mathrm{HPS}}$ and the GNS Hilbert space $\mathcal{H}_{\mathrm{DEHK}}$? *No* (Proposition 1), because their spectra have incompatible support.

The negative answer is sharper than a generic "algebra-type-mismatch" argument: it pinpoints the obstruction in a concrete, falsifiable way (bounded versus unbounded continuous spectrum). The thermodynamic match that survives, $\log Z_{\mathrm{chord}} = \log Z_{\mathrm{DEHK}} + O(1/N)$, is *not* new (it reduces to the Bekenstein–Hawking entropy of the cosmological horizon), and we therefore frame this paper at the level of a consistency note: a clean lemma (Theorem 1) plus a clean obstruction (Proposition 1) plus an honest account of what is left open.

Plan. Section 2 states and proves the Araki-cocycle identity $dS_{\mathrm{gen}}/d\tau_R = \langle K_R \rangle_\rho$ in type-II_∞. Section 3 reviews the modular-rigidified Stinespring embedding "C_{BS}" of the chord Hilbert space into the GNS Hilbert space of $\mathcal{M}_R$, and isolates the operator-level requirements (P1)–(P3) that any genuine bridge would have to satisfy. Section 4 states the spectrum-mismatch obstruction. Section 5 records the Krylov-thermodynamic match that does survive. Section 6 discusses alternative routes (Faulkner–Hollands II type-changing recovery, modular Krylov density-of-states matches) and open problems.

2 Wk2: Araki cocycle in type-II$_\infty$

2.1 Setup

Let $\mathcal{M}$ be a von Neumann algebra in standard form on a Hilbert space $\mathcal{H}$, and let σ be a faithful normal state with GNS vector $|\Omega_\sigma\rangle$ in the natural positive cone $\mathcal{P}^\sharp$. For a second faithful normal state ρ with GNS vector $|\Omega_\rho\rangle \in \mathcal{P}^\sharp$, the *Araki relative modular operator* [7] is

$$\Delta_{\rho|\sigma} \;=\; S^*_{\rho|\sigma}\, S_{\rho|\sigma}, \qquad S_{\rho|\sigma} : A|\Omega_\sigma\rangle \longmapsto A^*|\Omega_\rho\rangle, \quad A \in \mathcal{M}. \tag{1}$$

Let σ_t^σ denote the modular automorphism group of σ on $\mathcal{M}$, generated by the modular Hamiltonian $K_R = -\log \Delta_\sigma \geq 0$ (uniform sign convention, cf. Wk2 memo [20]). The *Araki relative entropy* is

$$S(\rho\|\sigma) \;=\; -\,\langle\Omega_\rho|\, \log \Delta_{\rho|\sigma}\, |\Omega_\rho\rangle, \tag{2}$$

finite when $\mathrm{supp}(\rho) \subset \mathrm{supp}(\sigma)$ and $+\infty$ otherwise.

2.2 Type-II$_\infty$ extension via Pusz–Woronowicz

For the de Sitter crossed product $\mathcal{M}_R = \mathcal{A}_{\mathrm{III}_1} \rtimes_{\sigma^\phi} \mathbb{R}$ of CLPW [2] and DEHK [3], $\mathcal{M}_R$ is type-II$_\infty$ and admits a normal semifinite faithful (nsf) tracial weight τ_{II} (the Haagerup trace). The Araki construction is purely algebraic and the Tomita–Takesaki theory applies to any standard form, so $\Delta_{\rho|\sigma}$ is well-defined for every pair of nsf weights. The Pusz–Woronowicz [8] variational characterization

$$S(\rho\|\sigma) \;=\; \sup_{\substack{A\in\mathcal{M}\\ A=A^*\geq 0}} \big\{\, \rho(\log A) - \log \sigma(e^A) \,\big\} \tag{3}$$

extends to type-II$_\infty$ weights through Haagerup's L^p theory and the Ohya–Petz [10] formulation in semifinite algebras.

2.3 The cocycle identity and modular-time derivative

The relative modular operator satisfies the Connes-cocycle identity (a direct consequence of σ_t^σ acting on the GNS data):

$$\Delta_{\rho|\sigma\circ\sigma^\sigma_{-t}} \;=\; \Delta_{\rho|\sigma}\, e^{tK_R}, \tag{4}$$

in the strong-resolvent sense on the joint domain. (See Bratteli–Robinson [9], Theorem 5.3.10 for the analyticity strip $0 \leq \mathrm{Im}\, t \leq \beta/2$ of Δ_σ^{it} on KMS states, and Theorem 5.3.15 for the differentiability of relative entropy along the modular flow.)

Theorem 1 (Araki rate, Wk2)**.** *Let $\mathcal{M}_R$ be a type-II$_\infty$ crossed-product algebra with nsf weight τ_{II}, KMS reference state σ at inverse temperature β_{dS} with modular Hamiltonian K_R, and let ρ be a normal state with* $\mathrm{supp}(\rho) \subset \mathrm{supp}(\sigma)$ *satisfying the regularity condition $\rho \in \mathrm{dom}\, K_R$ (i.e. $\langle K_R\rangle_\rho < \infty$). Define the modular-time-evolved reference $\sigma_{\tau_R} = \sigma \circ \sigma^\sigma_{-\tau_R}$. Then*

$$\frac{d}{d\tau_R}\, S\big(\rho\|\sigma_{\tau_R}\big)\Big|_{\tau_R=0} \;=\; -\,\langle K_R\rangle_\rho, \tag{5}$$

and consequently, with the FS24 identification $S_{\rm gen}(\rho) = S_{\rm gen}(\sigma) - S(\rho\|\sigma) +$ (KMS boundary terms) [4],

$$\boxed{\left.\frac{dS_{\rm gen}}{d\tau_R}\right|_{\rho\ \textit{fixed}} \;=\; +\langle K_R\rangle_\rho.} \tag{6}$$

Proof. By (4), $\log\Delta_{\rho|\sigma_{\tau_R}} = \log\Delta_{\rho|\sigma} + \tau_R K_R$ on the joint analyticity domain. Differentiating the formula (2), exchanging the $d/d\tau_R$ with the GNS expectation (justified by the dominated convergence on the strip via the KMS analyticity of Bratteli–Robinson [9], Thm. 5.3.10), one obtains

$$\left.\frac{d}{d\tau_R} S(\rho\|\sigma_{\tau_R})\right|_{\tau_R=0} \;=\; -\langle\Omega_\rho|\, K_R\, |\Omega_\rho\rangle \;=\; -\langle K_R\rangle_\rho,$$

which is (5). The conversion to $S_{\rm gen}$ through the FS24 identification gives (6); the boundary terms vanish for the KMS reference σ on a Killing horizon (FS24, Sec. 3). □

Remark 1. *The regularity hypothesis $\rho \in \operatorname{dom} K_R$ is non-trivial in type-II$_\infty$: it requires ρ to lie in the form-domain of K_R, equivalent to a finite* modular energy *expectation. For Boulware or perturbed-thermal reference states in the SdS static patch, this condition is plausible but has not been verified state-by-state in the DEHK literature. We treat* (6) *as a conditional theorem in this sense.*

Remark 2 (Type-I sanity check)**.** *In the type-I limit (replace $\mathcal{M}_R$ by $\mathcal{B}(\mathcal{H})$ with $\sigma = \rho_\sigma \in \mathcal{S}_1(\mathcal{H})$ a density matrix), $\Delta_\sigma = \rho_\sigma \otimes \rho_\sigma^{-1}$ in the standard representation, $K_R = -\log\rho_\sigma$, and* (5) *reduces to the textbook formula $d/dt\, S(\rho\|\rho_\sigma e^{-tK_R})|_{t=0} = -\mathrm{Tr}(\rho K_R)$ of Uhlmann [11] and Lindblad [12]. The type-II$_\infty$ statement is the natural extension to a setting where ρ_σ is not trace-class.*

Compatibility with the FS24 GSL. Equation (6) is consistent with the Faulkner–Speranza GSL [4]: monotonicity of relative entropy under the modular inclusions of $\mathcal{M}_R$ along the horizon implies $d/d\tau_R\, S(\rho\|\sigma) \geq 0$ for cuts moving into the future (modular flow as "algebraic time"), and Theorem 1 identifies the rate with $\langle K_R\rangle_\rho$. This is the algebraic analogue of Wall's GSL theorem [13] as recast in the crossed-product language.

3 Wk3: Modular-rigidified Stinespring embedding

We now turn to the question whether K_R admits a holographic dual identification with the chord Hamiltonian of the DSSYK / sine-dilaton holographic dictionary.

3.1 The chord-side data (HPS / HOPSW)

HPS [5] construct, from the chord Hilbert space $\mathcal{H}_{\rm HPS} = \ell^2(\mathbb{N}_0)$ with chord-creation operator $a^\dagger$ and chord-number basis $|n\rangle = (a^\dagger)^n|0\rangle/\sqrt{[n]_q!}$, the gravitational transfer-matrix Hamiltonian

$$H_{\rm grav} \;=\; \frac{1}{\sqrt{2|\log q|(1-q^2)}}\Big[-\cos(\hat{P}) + \tfrac{1}{2}e^{i\hat{P}}e^{-\hat{L}}\Big], \tag{7}$$

acting on $\mathcal{H}_{\rm HPS}$, where $\hat{L} = 2|\log q|\hat{n}$ is the chord-length operator and $\hat{P}$ its conjugate. The spectrum is

$$\mathrm{spec}(H_{\rm grav}) \;=\; \Big\{\, E(\theta) = -\cos\theta\,\big[2|\log q|(1-q^2)\big]^{-1/2} : \theta \in [0,\pi] \,\Big\}, \tag{8}$$

which is bounded continuous on $[-E_{\max}, +E_{\max}]$ with $E_{\max} = [2|\log q|(1-q^2)]^{-1/2}$. The Plancherel measure is $d\mu(\theta) = (q^2, e^{\pm 2i\theta}; q^2)_\infty \, d\theta$ (HPS, eq. 28).

HOPSW [6] establish the holographic dictionary $\hat{L} \leftrightarrow$ extremal timelike volume in the dS_2 geometry anchored at past/future infinities, and identify $H_{\rm grav}$ with the ADM Hamiltonian of sine-dilaton dS_2 in the high-energy $\theta \to \pi$ limit. The Hawking temperature reads $\beta_{\rm dS}(\theta) = -2\pi/\theta$ in the dS regime (HOPSW, eq. (17) and following).

3.2 The crossed-product side (CLPW / DEHK)

On the gravity side, CLPW [2] and DEHK [3] construct the crossed-product algebra $\mathcal{M}_R = \mathcal{A}_R \rtimes_{\sigma^\phi} \mathbb{R}$ acting on $\mathcal{H}_{\rm DEHK} = \mathcal{H}_{\rm qft} \otimes L^2(\mathbb{R})$, generated by $\mathcal{A}_R$ (the bulk QFT type-III_1 algebra in the static patch), the dressed observables $e^{ipH_{\rm mod}}\, a\, e^{-ipH_{\rm mod}}$, and the clock momentum p. The dressed modular Hamiltonian is (CLPW, eq. (2.10) footnote; DEHK, eq. (3.6))

$$K_R \;=\; \beta_{\rm dS}\, H_\xi \;+\; p, \tag{9}$$

where H_ξ is the boost generator (continuous unbounded spectrum on $\mathbb{R}$ since $\mathcal{A}_R$ is type-III_1) and p is the clock momentum (continuous spectrum on $\mathbb{R}$ for an ideal clock). Hence

$$\mathrm{spec}(K_R) \;=\; \mathbb{R}, \tag{10}$$

unbounded both above and below.

3.3 The modular-rigidified Stinespring embedding C_{BS}

The natural attempt to bridge the two sides is a CPTP isometry $\Phi : \mathcal{B}(\mathcal{H}_{\rm HPS}) \to \mathcal{M}_R$ obtained from a Stinespring isometry $V : \mathcal{H}_{\rm HPS} \to \mathcal{H}_{\rm DEHK}$ with $V|n\rangle = |\xi_n\rangle$, where $\{|\xi_n\rangle\}_{n\geq 0}$ are analytic vectors in the natural positive cone $\mathcal{P}^\sharp$ of $(\mathcal{M}_R, \sigma)$ chosen so as to be eigenstates of $\hat{L}$ under the HOPSW dictionary. Berry-phase ambiguity in V is rigidified by the Caputa–Magán–Patramanis–Tonni [16] modular-Lyapunov universality $\lambda_L^{(\rm mod)} = 2\pi$ together with the Connes cocycle compatibility, fixing V up to a global phase.

The bridge would qualify as a *genuine* algebraic identification if it satisfied:

(P1) (state matching) $\Phi(\rho_{\rm HPS}) = \rho_{\rm DEHK}$,

(P2) (modular-flow intertwining) $\Phi \circ \sigma_t^{(\rm HPS)} = \sigma_t^{(\rm DEHK)} \circ \Phi$ for all $t \in \mathbb{R}$,

(P3) (Connes-cocycle equivariance) the cocycle $[D\rho_{\rm DEHK} : D\sigma_{\rm DEHK}]_t \in \mathcal{M}_R$ pulls back to a corresponding cocycle on $\mathcal{B}(\mathcal{H}_{\rm HPS})$.

The detailed construction of V, the Stinespring map $\Phi(B) = \pi_R(VBV^*)$, and the Berry-rigidification by the cocycle identity are reviewed in the predecessor memo [21]. What concerns us here is whether (P1)–(P3) are operator-level attainable.

4 Wk4: Spectrum-mismatch no-go

Proposition 1 (No-go for HPS–DEHK Stinespring intertwining)**.** *Let H_{grav} be the chord (transfer-matrix) Hamiltonian on $\mathcal{H}_{\text{HPS}}$ with spectrum* (8) *bounded on $[-E_{\max}, +E_{\max}]$, and let K_R be the dressed modular Hamiltonian on $\mathcal{H}_{\text{DEHK}}$ with spectrum $\mathbb{R}$. Then no isometry $V : \mathcal{H}_{\text{HPS}} \to \mathcal{H}_{\text{DEHK}}$ can intertwine H_{grav} with K_R in the sense of $V\, H_{\text{grav}} = K_R\, V$ on $\operatorname{dom} H_{\text{grav}}$. Consequently, the modular-rigidified Stinespring embedding $\Phi : \mathcal{B}(\mathcal{H}_{\text{HPS}}) \to \mathcal{M}_R$ of Section 3 cannot satisfy (P2) in any non-trivial sense, and (P1), (P3) fail accordingly.*

Proof. Suppose, for contradiction, that $V\, H_{\text{grav}} = K_R\, V$ on a core for H_{grav}. Since V is an isometry, VV^* is the orthogonal projection onto $\operatorname{ran} V \subset \mathcal{H}_{\text{DEHK}}$, and $V H_{\text{grav}} V^* = K_R\, VV^*$. Then for any unit vector $\psi \in \operatorname{dom} H_{\text{grav}}$,

$$\langle V\psi |\, K_R\, | V\psi \rangle \;=\; \langle \psi |\, H_{\text{grav}}\, | \psi \rangle \;\in\; [-E_{\max}, +E_{\max}]$$

by (8). Hence $\operatorname{ran} V$ would lie in the spectral subspace $\chi_{[-E_{\max},+E_{\max}]}(K_R)\, \mathcal{H}_{\text{DEHK}}$, i.e. K_R restricted to $\operatorname{ran} V$ would be bounded. But $\operatorname{ran} V$ is infinite-dimensional (since V is an isometry from $\ell^2(\mathbb{N}_0)$), and the restriction of a self-adjoint operator with continuous spectrum on $\mathbb{R}$ to any infinite-dimensional invariant subspace $\mathcal{K}$ either inherits unbounded spectrum (if $\mathcal{K}$ is not contained in a finite-spectral band) or is unitarily equivalent to a direct sum of bounded operators on spectral subspaces (if $\mathcal{K}$ *is* contained in a bounded spectral band of K_R). In the second case, $\operatorname{ran} V$ lives in $\chi_{[-E_{\max},+E_{\max}]}(K_R)\, \mathcal{H}_{\text{DEHK}}$, which is proper and inconsistent with V being an isometry whose image is modular-flow invariant under $\sigma_t^{(\text{DEHK})} = e^{itK_R} \cdot e^{-itK_R}$ (since unbounded K_R generates a flow that escapes any bounded spectral band on a generic $\operatorname{ran} V$ direction). The contradiction forces (P2) to fail.

(P1) fails because the β_{dS}-KMS state for K_R in $\mathcal{M}_R$ is supported on the full real spectrum, while $\rho_{\text{HPS}} = \sum_n p_n(\beta) |n\rangle\langle n|$ has support in the discrete chord basis with thermal weights determined by E_n^{chord} from the bounded spectrum (8); $\Phi(\rho_{\text{HPS}})$ then has bounded modular energy support $[-E_{\max}, +E_{\max}]$ inside $\mathcal{M}_R$, while the genuine $\rho_{\text{DEHK}} = e^{-\beta_{\text{dS}} K_R}/Z_{\text{DEHK}}$ does not. (P3) is a corollary, since the Connes cocycle is built from the relative modular operator $\Delta_{\rho_{\text{DEHK}}|\sigma_{\text{DEHK}}}$, which in turn requires (P1). □

Remark 3 (Categorical character of the obstruction)**.** *The obstruction is not a small-parameter / loop-correction artifact: it is* topological *on the spectrum. Even with arbitrarily small $|\log q|$, $E_{\max} \to \infty$ but $E(\theta)$ remains bounded for each fixed $q \in (0,1)$, while K_R is unbounded on $\mathbb{R}$ for every $\beta_{\text{dS}} > 0$. The two spectra never agree as sets.*

Remark 4 (On the type-I vs type-II$_\infty$ algebra mismatch)**.** *A coarser way to state the obstruction is that $\mathcal{B}(\mathcal{H}_{\text{HPS}})$ is type-I (the chord algebra is finite-dimensional at any chord-truncation level N_{chord} and infinite-dimensional but separable in the limit; its von Neumann completion is $\mathcal{B}(\mathcal{H}_{\text{HPS}})$, type-I), while $\mathcal{M}_R$ is type-II$_\infty$. Type-changing CPTP maps between general von Neumann algebras do exist (Faulkner–Hollands II [19]) and even admit FH-II approximate recovery channels under small relative-entropy perturbations. What Proposition 1 shows is that the* spectral obstruction *is sharper than the algebra-type obstruction: even within the type-changing FH-II framework, the candidate spectra $E(\theta)$ and* $\operatorname{spec}(K_R)$ *are incompatible.*

5 What survives: Krylov-thermodynamic match

The negative content of Section 4 is partial: while no operator-level isometry intertwines H_{grav} with K_R, the *thermodynamic* content of the two systems agrees at semiclassical order.

Proposition 2 (Krylov–thermodynamic match)**.** *In the regime $q \to 1^-$, $\theta \to \pi$ (the dS large-N regime of HPS–HOPSW), the partition functions $Z_{\text{chord}}(\beta) := \int_0^\pi (q^2, e^{\pm 2i\theta}; q^2)_\infty \, e^{-\beta E(\theta)} \, d\theta$ and $Z_{\text{DEHK}}(\beta) := \tau_{\text{II}}(e^{-\beta K_R})$ satisfy, at the de Sitter inverse temperature β_{dS}:*

$$\log Z_{\text{chord}}(\beta_{\text{dS}}) \;=\; \log Z_{\text{DEHK}}(\beta_{\text{dS}}) \;+\; O(1/N), \tag{11}$$

both reproducing the Bekenstein–Hawking entropy $S_{\text{BH}} = A_{\text{horizon}}/(4G_N)$ at leading order.

Sketch. This is essentially a re-statement of (i) the standard Bekenstein–Hawking entropy of the cosmological horizon in CLPW/DEHK at leading $1/G_N$ [2, 3], and (ii) HOPSW eq. (27) (chord-Krylov disk partition function $\to$ dS_2 Bekenstein–Hawking entropy) [6], both regulated through the same β_{dS}-KMS thermal data on the cosmological horizon. At higher orders, the $1/N$ corrections differ: chord-side corrections follow from the q-Hermite Plancherel measure expansion; DEHK-side corrections follow from the loop expansion around the saddle of $\tau_{\text{II}}(e^{-\beta K_R})$. We do not claim agreement beyond leading order. □

Remark 5. *Proposition 2 is* not *a new result: the Bekenstein–Hawking entropy of the cosmological horizon is recovered by* every *consistent dS holography proposal at semiclassical order. We include it here only to record* exactly *what part of the attempted bridge survives the no-go in Proposition 1: the partition-function thermodynamics, but not the underlying operator content.*

Remark 6 (Krylov-asymptotic match)**.** *A second weaker statement that survives is a Lanczos-asymptotic match. The HPS chord algebra has Lanczos coefficients $b_n = \sqrt{[n]_q(1-q^n)} \sim \alpha n + \gamma$ at large n (linear growth, signature of $\lambda_L = 2\pi T$). For K_R, Vardian's modular Krylov procedure [17] (in an AdS/CFT setting; transfer to dS is by methodological analogy) yields modular Lanczos coefficients b_n^{mod} also growing linearly at late modular time with $\lambda_L^{(\text{mod})} = 2\pi$, the Caputa–Magán–Patramanis–Tonni universality [16]. Both spectral functions exhibit the same asymptotic Lyapunov exponent, but as different Jacobi matrices with different short-distance / low-n structure. This is a Krylov-asymptotic match, not an operator match, and is consistent with Proposition 1.*

6 Discussion and open problems

What is genuinely new. Theorem 1 is, as far as we are aware, the first explicit statement of the Araki-cocycle identity $dS_{\text{gen}}/d\tau_R = \langle K_R \rangle_\rho$ in the type-II_∞ crossed-product setting of CLPW/DEHK. It complements the FS24 GSL [4] by giving an explicit *rate* formula in terms of the modular-energy expectation, and makes the modular-time first law $\delta S_{\text{gen}} = \beta_{\text{dS}} \, \delta\langle K_R \rangle$ algebraically explicit on the type-II_∞ trace.

What is genuinely refuted. The natural attempt to identify $\langle K_R \rangle_\rho$ with chord-Krylov complexity C_K via a Stinespring isometry between the DSSYK chord Hilbert space and the GNS Hilbert space of $\mathcal{M}_R$ is refuted at the operator level by Proposition 1, because the spectra of the chord Hamiltonian and K_R have incompatible support (bounded vs

unbounded). What survives is the thermodynamic content of Proposition 2, i.e. the standard Bekenstein–Hawking dS entropy match, and the Krylov-asymptotic universal $\lambda_L = 2\pi$ shared between the two Lanczos sequences.

Alternative routes. Three logically distinct alternatives to the Stinespring intertwining route remain open:

1. *Type-changing FH-II recovery.* Faulkner–Hollands II [19] (and the predecessor FHSW [18]) construct universal approximate recovery channels for 2-positive maps between general von Neumann algebras, including the type-changing case. Their bound $S(\rho\|\sigma) - S(\Phi(\rho)\|\Phi(\sigma)) \geq -2\log F(\rho, R\circ\Phi(\rho))$ controls relative-entropy loss without requiring spectral matching of the modular Hamiltonians. A reformulation of the bridge problem in FH-II language might give an *approximate-recovery* statement that escapes the spectrum-mismatch no-go, but at the cost of giving up exact intertwining.

2. *Bounded function of K_R.* A bounded function $F(K_R)$ (e.g. $F(K_R) = (2/\pi)\arctan(\beta_{\mathrm{dS}} K_R/2)$) has spectrum in $(-1, +1)$ and could in principle match a rescaled H_{grav}. This is an ad hoc construction without independent physical motivation; we mention it for completeness only.

3. *Modular density-of-states matching.* A genuine result at the level of the spectral measure of K_R, namely that the chord-Krylov density of states $\rho(\theta)$ matches a suitable projection of the spectral measure of K_R in a controlled high-energy / late-modular-time limit, would yield a v6.2-style consistency note rather than a theorem. This route uses the CMPT [16] modular-Lyapunov universality and the modular Krylov approach of Vardian [17] (with the caveat that Vardian works in AdS/CFT and the dS extension requires an independent construction).

Closing assessment. The ER=EPR programme in the algebraic crossed-product setting yields, at the rigorous mathematical level, the Araki-cocycle rate formula (6) as a genuine new theorem (Theorem 1), and a clean operator-level no-go for the chord–K_R bridge (Proposition 1). The survival is the standard Bekenstein–Hawking thermodynamic match (Proposition 2), which was already known from independent considerations.

We do not claim that the ER=EPR / chord-vs-modular-complexity programme is closed, only that the most natural "Stinespring isometry" bridge is ruled out, and that future work must use either type-changing approximate recovery in the FH-II sense, a bounded-functional-calculus construction, or a coarser density-of-states / Lanczos-asymptotic matching at the level of spectral measures rather than operator identifications.

Acknowledgements

We thank the algebraic gravity community for the foundational results in [1–6] and the relative-entropy / Connes-cocycle community for the foundational results in [7–9, 11, 12]. The framing of this note as a "consistency note" rather than a full theorem reflects the obstructions identified in Sections 4, and we thank the predecessor portfolio (Wk1 Petz route, Wk2 Araki memo, Wk3 Berry–Stinespring memo, Wk4 spectrum memo) for sharpening the questions.

References

[1] E. Witten, "Gravity and the crossed product," *JHEP* **10** (2022) 008, doi:10.1007/JHEP10(2022)008, arXiv:2112.12828 [hep-th].

[2] V. Chandrasekaran, R. Longo, G. Penington, and E. Witten, "An algebra of observables for de Sitter space," *JHEP* **02** (2023) 082, doi:10.1007/JHEP02(2023)082, arXiv:2206.10780 [hep-th].

[3] J. De Vuyst, S. Eccles, P. A. Höhn, and J. Kirklin, "Crossed products and quantum reference frames: on the observer-dependence of gravitational entropy," *JHEP* **07** (2025) 063, doi:10.1007/JHEP07(2025)063, arXiv:2412.15502 [hep-th].

[4] T. Faulkner and A. J. Speranza, "Gravitational algebras and the generalized second law," *JHEP* **11** (2024) 099, doi:10.1007/JHEP11(2024)099, arXiv:2405.00847 [hep-th].

[5] M. P. Heller, J. Papalini, and T. Schuhmann, "Krylov spread complexity as holographic complexity beyond JT gravity," *Phys. Rev. Lett.* **135** (2025) 151602, doi:10.1103/PhysRevLett.135.151602, arXiv:2412.17785 [hep-th].

[6] M. P. Heller, F. Ori, J. Papalini, T. Schuhmann, and M.-T. Wang, "De Sitter holographic complexity from Krylov complexity in DSSYK," arXiv:2510.13986 [hep-th] (2025).

[7] H. Araki, "Relative entropy of states of von Neumann algebras," *Publ. RIMS, Kyoto Univ.* **11** (1976) 809–833; follow-up: *Comm. Math. Phys.* **66** (1977) 83.

[8] W. Pusz and S. L. Woronowicz, "Form convex functions and the WYDL and other inequalities," *Lett. Math. Phys.* **2** (1978) 505–512; related: *Rep. Math. Phys.* **8** (1975) 159.

[9] O. Bratteli and D. W. Robinson, *Operator Algebras and Quantum Statistical Mechanics, Vol. 2*, 2nd ed., Springer, 1997.

[10] M. Ohya and D. Petz, *Quantum Entropy and Its Use*, Springer, 1993.

[11] A. Uhlmann, "Relative entropy and the Wigner–Yanase–Dyson–Lieb concavity in an interpolation theory," *Comm. Math. Phys.* **54** (1977) 21.

[12] G. Lindblad, "Expectations and entropy inequalities for finite quantum systems," *Comm. Math. Phys.* **39** (1974) 111.

[13] A. C. Wall, "A proof of the generalized second law for rapidly changing fields and arbitrary horizon slices," *Phys. Rev. D* **85** (2012) 104049, doi:10.1103/PhysRevD.85.104049, arXiv:1105.3445 [gr-qc].

[14] J. J. Bisognano and E. H. Wichmann, "On the duality condition for a Hermitian scalar field," *J. Math. Phys.* **17** (1976) 303.

[15] A. Connes and C. Rovelli, "Von Neumann algebra automorphisms and time-thermodynamics relation in generally covariant quantum theories," *Class. Quantum Grav.* **11** (1994) 2899–2917, doi:10.1088/0264-9381/11/12/007.

[16] P. Caputa, J. M. Magán, D. Patramanis, and E. Tonni, "Krylov complexity of modular Hamiltonian evolution," *Phys. Rev. D* **109** (2024) 086004, doi:10.1103/PhysRevD.109.086004, arXiv:2306.14732 [hep-th].

[17] N. Vardian, "Modular Krylov complexity as a boundary probe of area operator and entanglement islands," arXiv:2602.02675 [hep-th] (2026).

[18] T. Faulkner, S. Hollands, B. Swingle, and Y. Wang, "Approximate recovery and relative entropy I: general von Neumann subalgebras," *Comm. Math. Phys.* **389** (2022) 349–397, arXiv:2006.08002 [hep-th].

[19] T. Faulkner and S. Hollands, "Approximate recoverability and relative entropy II: 2-positive channels of general von Neumann algebras," *Lett. Math. Phys.* **112** (2022) 26, arXiv:2010.05513 [hep-th].

[20] [ECI author(s)], "Araki Relative Modular Operator in Type-II$_\infty$: Week 2 Scoping Memo (ER=EPR Reopen, A13-2)," internal portfolio memo (2026-05-04).

[21] [ECI author(s)], "Berry–Stinespring channel C_{BS} for HPS (type-I) $\leftrightarrow$ DEHK (type-II$_\infty$): Week 3 memo (ER=EPR Reopen, A6)," internal portfolio memo (2026-05-05).